What the n_∞ lifetime fit determines, and the truncation bias on D_e/D_f

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Two questions came up in the n_∞ analysis. This note answers both from the data.

1. What does the global multi-fluence lifetime fit determine? It pins down three rates: how fast excitons recombine (ν), how fast carriers are trapped (λ), and how fast trapped charge decays (γ). It does not pin down the free-charge bimolecular rate ρ. The Saha equilibrium ties ρ to ν so tightly that the fit sees only their sum.
2. Does the measurement (truncation) bias distort the D_e/D_f fit? No. It is negligible against the D_e/D_f error bars.

All numbers come from the three n_∞ TPLM fluences, $N_{exc} = 5.5\times10^{17}, 3.1\times10^{18}, 7.2\times10^{19}\,\mathrm{cm}^{-3}$.

1 What the global lifetime fit determines

To learn which rates the three decays pin down, we fit all three PL curves at once with one shared rate set (ν, ρ, λ, γ, and the trap emission ratio q_{trap}, the share of the light that comes from trapped charge); only N_{exc} differs between the curves. This shared-rate fit reproduces the overall decay shape (Fig. 1). It under-fits the late-time tail at the two lower fluences. That tail constrains the trap parameters (γ and q_{trap}), not the result below, so we set it aside here.

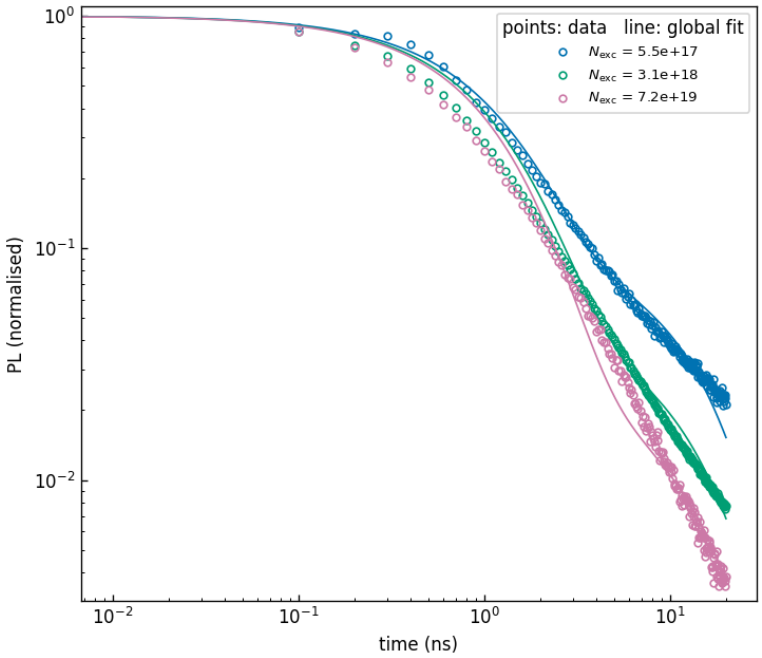


Figure 1: Global multi-fluence fit. Points: data; lines: the shared-rate fit. The overall decay is reproduced; the late-time tail at the two lower fluences is under-fit.

To find which rates the data determine, we test each rate one at a time. We fix the rate at a series of values; at each value we re-fit the other rates and measure how well the fit matches the data. That measure is the sum of squared residuals (SSR): for each data point we take the gap between the fitted curve and the data, square it, and add the squares over all points. (We use the logarithm of the PL, because the decay spans several decades, so we compare ratios rather than absolute differences.) A small SSR means a good fit. This recipe (fix one rate, re-fit the

rest, read the SSR) is the profile-likelihood method, the same method used for the D_e/D_f error bars. It uses no random sampling, so it returns the same answer every run.

We then read each rate from the shape of its SSR. A clear minimum means the rate is determined; a flat profile means it is not. The width of the minimum sets the error bar: a value is still acceptable as long as forcing the rate there barely worsens the fit, and "barely" is set by the size of the data's own scatter. That scatter is the residual variance, $s^2 = \mathrm{SSR}_{\min}/(N-p) = 4.2\times10^{-3}$, where $N = 600$ is the number of data points and $p = 6$ the number of fitted rates. The $\sim 1\sigma$ interval for a rate is the range over which its SSR rises by no more than s^2 above the minimum (Fig. 2).

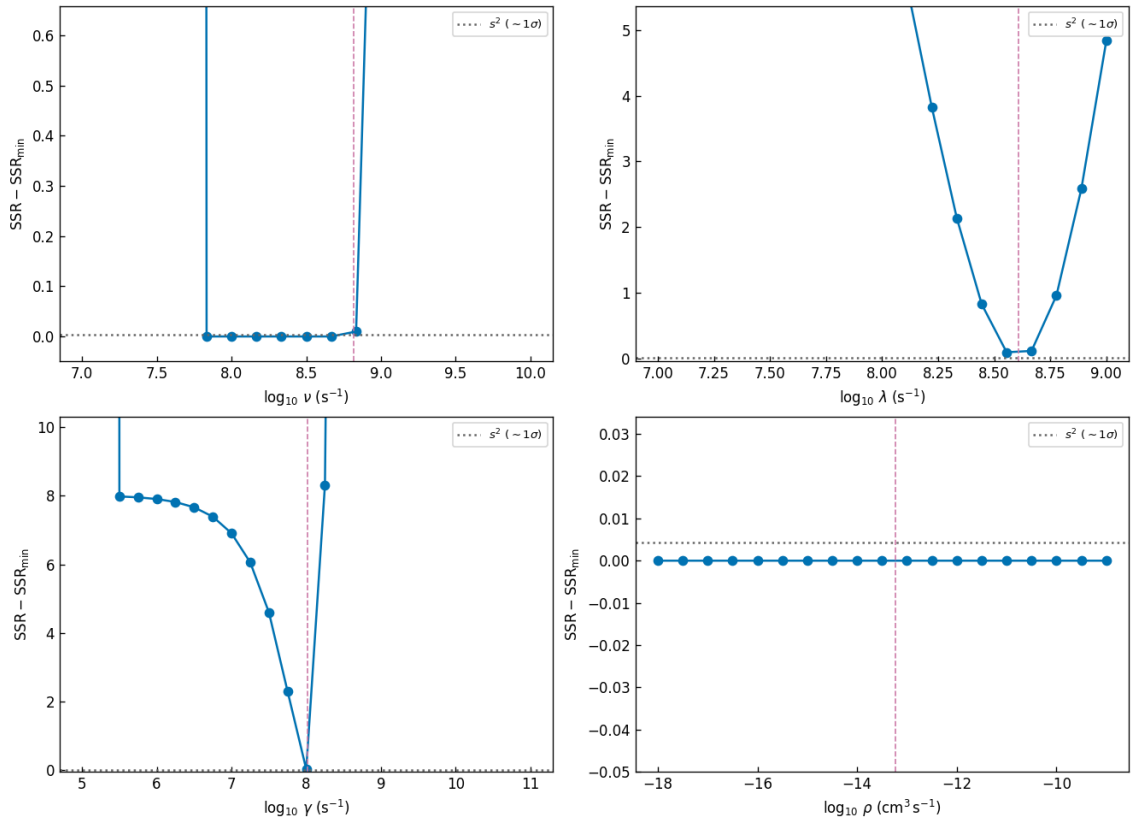


Figure 2: Profile likelihood of each rate (SSR minus its minimum, versus the value the rate is fixed at; dotted line: the s^2 threshold; dashed line: the best fit). The shapes tell the story: ν is a plateau with a hard lower wall, so it is bounded but not pinned to one value; λ is a clean well (well determined); γ has a minimum near $1 \times 10^8\,\mathrm{s}^{-1}$; ρ is flat, so the data do not determine it.

Table 1: What the three decays determine. Rates in s^{-1}; ρ in cm^3/s.

rate	meaning	result
ν	exciton recombination	determined as a bracket, $\sim [0.8, 6] \times 10^8$; rejects $> 10^9$
λ	carrier trapping	determined, 4.06×10^8, 1σ $[3.96, 4.17] \times 10^8$
γ	trapped-charge decay	determined, $\sim 1 \times 10^8$ (trap lifetime $\sim$10 ns)
ρ	free-charge bimolecular	**not determined** (flat profile, 10^{-18} to 10^{-9})

The results are in Table 1. ν is bounded, not pinned to a point. Its SSR is a plateau: equally good for any ν across a wide band, sharply worse below about $8 \times 10^7\,\mathrm{s}^{-1}$, and gradually worse above. So the data place a firm floor under ν and a soft ceiling over it, rejecting $\nu > 1 \times 10^9\,\mathrm{s}^{-1}$ and fencing it into roughly $[0.8, 6] \times 10^8\,\mathrm{s}^{-1}$. We call this a bracket: a range, not a single value.

λ is different. It has a clean minimum and is the best-determined rate: $\lambda = 4.06 \times 10^8\,\mathrm{s}^{-1}$, with a

formal 1σ interval $[3.96, 4.17] \times 10^8\,\mathrm{s}^{-1}$ (half-width 2.6%, from a fine grid around the minimum). One caution applies here above all. This interval assumes the residuals are independent. They are not: a smooth decay makes neighbouring residuals move together, so the true uncertainty on λ is wider than 2.6%.

Why the fit cannot determine ρ. The reason is exact, not a matter of weak signal, and it follows from the Saha equilibrium. Two rates are degenerate when the data cannot tell them apart: any increase in one can be cancelled by a decrease in the other, leaving the fitted curves identical, so the fit pins only their combination. Here that combination is $\nu + \rho A$, and the cause is a single algebraic fact.

The Saha free-charge fraction $x = 2/\left(\sqrt{4n/A + 1} + 1\right)$ obeys the identity

$$1 - x = x^2\, n/A, \tag{1}$$

at every density (we checked this by algebra, and numerically to 6×10^{-14}). Here A is the Saha density scale, the density at which carriers are half excitons and half free charge. The identity makes the two recombination channels share one density shape: the exciton per-carrier rate $\nu(1 - x)$ equals $\nu\, x^2 n/A$, a fixed multiple of the bimolecular per-carrier rate $\rho\, x^2 n$, with the ratio $\nu/(\rho A)$ constant at all densities (Fig. 3). The total recombination then collapses to one term,

$$\nu(1 - x) + \rho\, x^2 n = x^2 n\, \left(\nu/A + \rho\right), \tag{2}$$

so the data see only the sum $\nu + \rho A$. Changing the fluence cannot separate ν from ρ, because the Saha coupling gives the two channels the same dependence on density.

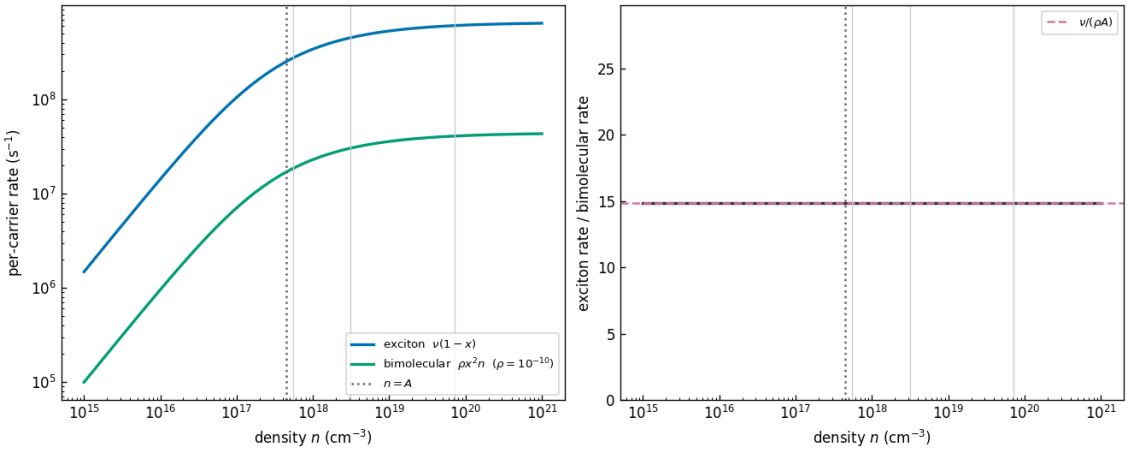


Figure 3: The Saha exact proportionality. Left: the exciton and bimolecular per-carrier rates versus density (parallel on the log axes). Right: their ratio, flat at $\nu/(\rho A)$ across all densities. Dotted line: $n = A$; light lines: the three measured fluences. The fit sees only $\nu + \rho A$.

The fit shows this directly (Fig. 4). When we fix ρ at a series of values and re-fit the rest, ν tracks $\nu_{\mathrm{eff}} - \rho A$ to four significant figures, the fit quality does not change, and λ and γ do not move (they shift by less than 0.01%). The fit holds $\nu + \rho A$ fixed and lets ν absorb the change. This continues until ρA reaches ν (near $\rho = 1.5 \times 10^{-9}\,\mathrm{cm}^3/\mathrm{s}$). There ν can no longer compensate, and the fit breaks. Only ν trades against ρ; the trap channel is not involved.

Free charges are not absent: the free-charge fraction at $t = 0$ is 0.58, 0.31, and 0.075 across the three fluences. ρ is invisible not because there are no free charges, but because the Saha coupling makes its channel rise and fall with density in step with the exciton channel. To separate the two, you need data where they scale differently. One option is the low-density regime $n \ll A$: there almost all charge is free ($x \to 1$), so the exciton channel vanishes and the bimolecular channel

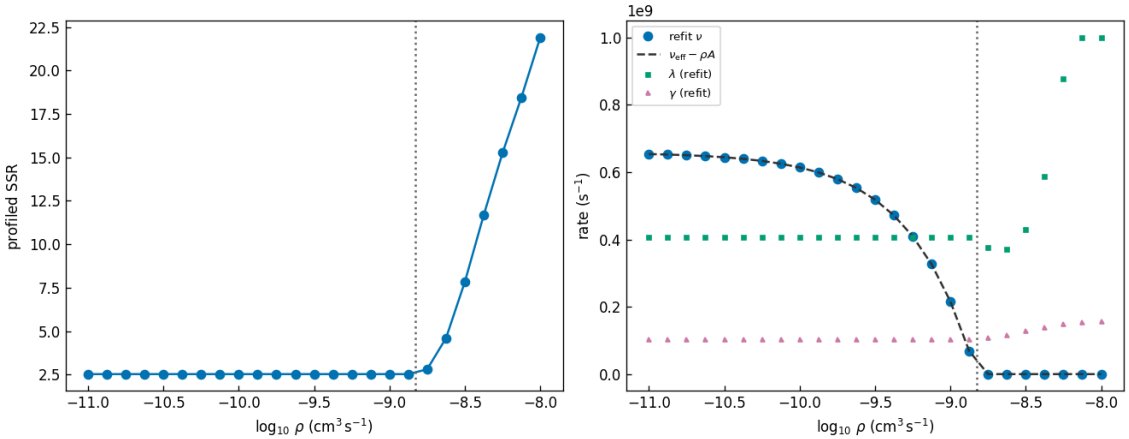


Figure 4: The ρ–ν trade. Left: the fit quality (SSR) is flat until ρA reaches ν (dotted line). Right: the re-fit ν tracks $\nu_{\text{eff}} - \rho A$ (dashed) while λ and γ stay fixed. The data determine $\nu + \rho A$, not ρ.

returns to $\sim \rho n$, and the two no longer track each other. Another is an independent measurement of the free charges, such as their diffusion D_{f}.

2 The truncation bias does not affect $D_{\text{e}}/D_{\text{f}}$

The width fit cannot return a negative width. So when the noise is large, the low side of the fitted-width scatter is cut off and only the high side survives, which pulls the average width upward. At low contrast-to-noise (CNR) the fitted width is therefore biased high, and a width that is too large inflates the late-time MSD. The question is whether that inflated MSD also inflates $D_{\text{e}}/D_{\text{f}}$.

It does not. The reason is a single mechanism: low CNR causes both the bias and a low fit weight. The $D_{\text{e}}/D_{\text{f}}$ fit weights each frame by how trustworthy it is, so a clean frame counts a lot and a noisy frame counts little (this is inverse-variance weighting, with weight $\propto 1/\sigma^2$ and σ the per-frame MSD uncertainty). A low-CNR frame is at once biased (from the truncation) and noisy (large σ), so the very frames that carry the bias barely count. These are not two independent safeguards; they are one fact.

Table 2 puts numbers on this. For each frame it lists the per-frame CNR (measured with fft-cnr on the recorded profiles) next to that frame's weight in the fit, $w_i = 1/\sigma_i^2$, with the weights scaled to sum to one. The bias is significant only near CNR 1–2. In the two dimmer fluences, a quarter to two-fifths of the frames fall in that range, but their summed weight is below 2×10^{-5} of the total. The weight-weighted mean CNR is 63–268, so the fit is carried by high-CNR frames, where there is no bias.

Table 2: Per-frame CNR and fit weight for the diffusion ($D_{\text{e}}/D_{\text{f}}$) fit. The bias-prone frames (CNR < 5) carry a vanishing share of the fit weight.

fluence (μJ/cm^2)	CNR range	frames with CNR < 5	weight share of those frames	weighted-mean CNR
13.11	5.4–336	none	—	268
2.298	1.8–137	$\sim 1/4$	1.9×10^{-5}	85
301.4	1.2–93	$\sim 2/5$	1.6×10^{-5}	63

A simple bound follows. In a weighted average, a group of points that holds weight fraction w can move the result by at most w times how wrong it is. The bias-prone frames here hold $w < 2 \times 10^{-5}$. Even if their widths were wrong by an amount as large as the MSD itself (a

deliberate worst case), they would move D by less than 2×10^{-5} of its value. The reported $D_\mathrm{e} = (0.030 \pm 0.026)\,\mathrm{cm}^2/\mathrm{s}$ has a relative error near 0.87. So the shift from the bias is about 4×10^4 times smaller than the error bar.

No truncation correction is needed on the diffusion fits. The limit on $D_\mathrm{e}/D_\mathrm{f}$ is the degeneracy between D_e and D_f (the large error bars), which the profile-likelihood error analysis addresses, not the bias.

3 What to report

- Report ν, λ, and γ from the lifetime fit; the data determine them.
- Do not fit ρ from the lifetime data. Fix it from the diffusion data or the literature, or report it as unconstrained by the lifetime. Once ρ is fixed, the degeneracy is broken and ν is determined, so you can leave ν free rather than fixing it too.
- Do not add a Meitner-Auger term ($-C\,n^3$). We tested it: its coefficient runs to zero and the fit quality does not change with it on or off, so the lifetime decays do not constrain it (the same kind of degeneracy as ρ). Keep its bound near the literature value ($C \sim 1{\times}10^{-28}\,\mathrm{cm}^6/\mathrm{s}$ for a lead halide), but do not trust a fitted value that sits at a bound or whose profile is flat.
- Choose the bounds for λ, γ, and C from physical ranges, wide enough that the best value sits inside. After fitting, check that no parameter sits at its bound: a value pinned at a bound is set by the bound, not the data. γ showed this. A bound placed at its best value made γ look pinned, and widening the bound fixed it.

Method. Lifetime identifiability: the shared-rate global fit, then a profile of each rate (fix one, re-fit the rest with a local optimiser, read the SSR against the fixed rate). Bias: the per-frame CNR from fft-cnr on the recorded profiles, against the $D_\mathrm{e}/D_\mathrm{f}$ fit weight $1/\sigma_i^2$ (the fit minimises $\chi^2 = \sum_i [(\mathrm{MSD}_i^{\mathrm{sim}} - \mathrm{MSD}_i)/\sigma_i]^2$). The analysis script (`lifetime_identifiability.py`) regenerates every figure here.